

Rhenium

75
Re

Liquid rhenium will not boil until it is at 5,590°C (10,094°F), which is about the same temperature as the surface of the Sun. This metal has the highest boiling point of any element. Rhenium is used to make heat-resistant superalloys for X-ray machines and the exhaust nozzles for jet fighters and rockets.

ATOMIC MASS 186.207

STATE Solid

DISCOVERY 1925 (Walter Noddack, Ida Tacke, and Otto Berg)

Iridium

77
Ir

One of Earth's rarest elements, iridium is found in a thin layer of clay in the planet's crust, and is especially seen in the Badlands of South Dakota, USA. Scientists think it was distributed by the meteorite that made the dinosaurs extinct.

ATOMIC MASS 192.217

STATE Solid

DISCOVERY 1803 (Smithson Tennant)



Osmium

76
Os

One cubic cm (0.06 cubic in) of osmium is more than 22 times heavier than the same amount of water. This transition metal is the densest element in the world. It can be used by police investigators to find clues at a crime scene. One of its compounds, osmium oxide, mixes well with oil so it is used in the search for fingerprints.

**Pellet of pure osmium
refined in a laboratory**

ATOMIC MASS 190.23

STATE Solid

DISCOVERY 1803
(Smithson Tennant)

